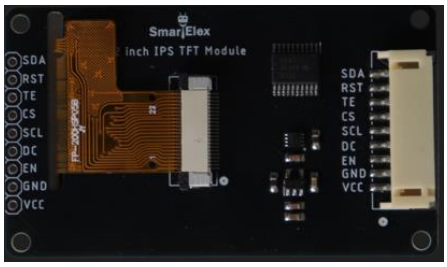
SmartElex 2 inch IPS TFT LCD module





❖ Introduction –

The **2.0 Inch TFT LCD Display Module (LHS200KP-IF05)** is a compact and high-performance color display featuring a **240×320 resolution** with **262K colors**. It is based on the **ST7789P3 driver IC**, making it suitable for embedded systems, Arduino, Raspberry Pi, and industrial applications.

This module supports both **4-wire SPI**, providing flexibility in interfacing. With **wide viewing angles**, high brightness, and a built-in **white LED backlight**, it is ideal for handheld devices, IoT projects, and HMI applications.

❖ Specifications –

- Display Size - 2.0 inch
- Resolution - 240 (H) · 320 (V)
- Display Mode - Normally Black
- Color Depth - 262K colors
- Driver IC - ST7789P3
- Interface - SPI / 8-bit MCU
- Pixel Pitch - 0.1275 · 0.1275 mm
- Active Area - 30.6 · 40.8 mm
- Module Size - 34.6 · 47.8 · 2.05 mm
- Viewing Direction - All viewing angle
- Brightness - 400 cd/m″ (typ.)
- Contrast Ratio - 640:1 (typ.)
- Response Time - 30-40 ms

❖ Features –

- 2.0-inch TFT LCD display
- Resolution: 240 × 320 pixels (RGB)
- 262K color support
- Interface: 4-line SPI
- Driver IC: ST7789P3
- High brightness: 400 cd/m² (typical)
- Wide viewing angle (All-direction view)

- Built-in white LED backlight (3 LEDs)
- Wide operating temperature range --20℃~ + 70℃

❖ Pin Connections –

Display Module	Arduino UNO
VCC	5V
GND	GND
EN	D6
DC	D9
SCL	SCL(A5)
CS	D10
TE	-
RST	D8
SDA	SDA(A4)

❖ Handling Precautions -

- Avoid mechanical shock (glass panel)
- Do not press display surface
- Avoid static electricity (use grounding)
- Do not expose to solvents or moisture
- Do not disassemble module

❖ Applications -

- Arduino & Raspberry Pi Projects
- Industrial Display Panels
- Handheld Devices
- IoT Interfaces
- Embedded GUI Systems